

RISHI KIRAN KARNATAKAM

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Summary

Motivated Computer Science student seeking job and internship opportunities to apply my skills in software development and problem-solving. Quick to learn new technologies and adapt to challenges, focusing on contributing to innovative and impactful projects.

Education

- GPA: 8.4

- GPA: 9.4

- GPA: 9.8

Skills

- **Programming Languages:** C, Python
- **Databases:** MySQL, MongoDB
- **Cloud Platforms:** Amazon Web Services (AWS)
- **Machine Learning:** TensorFlow
- **Version Control:** Git, Github
- **Frameworks/Tools:** Flask, Jenkins

Projects

- Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.
- Improved disease detection accuracy by 25% using transfer learning with pre-trained CNN models.
- Achieved 90%+ accuracy on test datasets, optimizing performance for agricultural disease diagnosis.
- **Technologies:** Python, TensorFlow

- Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.
- Enhanced strategic depth and move accuracy by 20%.

- Achieved a 30% improvement in move generation speed through alpha-beta pruning.
- **Technologies:** Python, Tkinter, Chess Library

- Built a mobile app for real-time plant disease detection using camera or gallery images.
- Integrated a TensorFlow Lite model (Inception V3) for accurate disease prediction with offline support.
- Enabled local data storage and automatic syncing to MongoDB Atlas when connected online.
- Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.
- **Technologies:** Flutter, Dart, TensorFlow Lite, MongoDB Atlas

- Developed a 3D interactive solar system model to enhance gamified learning.
- Created a first-person 3D game inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.
- **Technologies:** C#, Unity Engine

Awards

- **Internal Hackathon on Generative AI** (Aug 2024): Achieved 3rd place in a college-hosted Hackathon focused on Generative AI.
- **National Hackathon on Generative AI** (Sep 2024): Achieved 2nd place in a National Level Hackathon focused on Generative AI.
- **National Hackathon on AR/VR Development** (Oct 2023): Ranked in the top 10 at a national-level hackathon focused on AR/VR.

Certifications

- **Supervised Machine Learning: Regression and Classification** (Jun 2023) – Stanford University (Coursera)
- **The Joy of Computing Using Python** (Jul 2023) – IIT Madras (NPTEL)
- **Python For Data Science** (Jan 2025) – IIT Madras (NPTEL)
- **AICTE Virtual Internship** (2024) – Nalla Narasimha Reddy Education Society's Group of Institutions
- **AWS Academy Graduate - AWS Academy Cloud Architecting** (Sep 2024) – AWS Academy